Subject: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2018 – Paper 3 – Solutions**

June 2018 – Question 1 – Solution

(a) Two propositions, $\boldsymbol{r}$ and $\boldsymbol{s}$, are given as

$\boldsymbol{r}$: It is Monday

$\boldsymbol{s}$: It is raining.

Express the following statements in symbolic form:

(i) It is Monday and it is raining.

In symbolic form: $\boldsymbol{r} \wedge \boldsymbol{s}$

(ii) It is not Monday or it is not raining. [2]

In symbolic form: $\sim\boldsymbol{r} \vee \sim\boldsymbol{s}$

(b) (i) Given that $\boldsymbol{p}$ and $\boldsymbol{q}$ are two propositions, complete the truth table below.

The completed table is shown below:

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{p}$	$\sim\boldsymbol{q}$	$\boldsymbol{p} \vee \boldsymbol{q}$	$\sim(\boldsymbol{p}) \wedge \sim(\boldsymbol{q})$	$(\boldsymbol{p} \vee \boldsymbol{q}) \wedge [\sim(\boldsymbol{p}) \wedge \sim\boldsymbol{q})]$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

(ii) Hence, state whether $(\boldsymbol{p} \vee \boldsymbol{q}) \wedge [\sim(\boldsymbol{p}) \wedge \sim\boldsymbol{q})]$ is a tautology or a contradiction.

Justify your response. [2]

The truth table shows that all truth values of $(\boldsymbol{p} \vee \boldsymbol{q}) \wedge [\sim(\boldsymbol{p}) \wedge \sim\boldsymbol{q})]$ is false.

Hence, it is a contradiction.

(c) Prove by mathematical induction that $2^{n+1} + 3^{2n-1}$ is a multiple of 7 for all positive integers $n \geq 1$. [7]

Let $P(n)$ be the statement:

$2^{n+1} + 3^{2n-1}$ is a multiple of 7 for all $n \in \mathbb{Z}^+, n \geq 1$.

Looking at $P(1)$.

$$\begin{aligned} 2^{(1)+1} + 3^{2(1)-1} &= 2^2 + 3^{2-1} \\ &= 4 + 3^1 \\ &= 4 + 3 \\ &= 7 \text{ which is a multiple of 7.} \end{aligned}$$

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $2^{k+1} + 3^{2k-1}$ is a multiple of 7.

So, we have,

$2^{k+1} + 3^{2k-1} = 7m$, where $m \in \mathbb{N}$

Looking at $P(k+1)$.

That is, $2^{k+2} + 3^{2k+1}$ is a multiple of 7.

We need to prove $P(k+1)$.

$$
\begin{aligned}
2^{k+2} + 3^{2k+1} &= 2^1 \cdot 2^{k+1} + 3^2 \cdot 3^{2k-1} \\
&= 2 \cdot 2^{k+1} + 9 \cdot 3^{2k-1} \\
&= 2 \cdot 2^{k+1} + (2+7) \cdot 3^{2k-1} \\
&= 2 \cdot 2^{k+1} + 2 \cdot 3^{2k-1} + 7 \cdot 3^{2k-1} \\
&= 2(2^{k+1} + 3^{2k-1}) + 7 \cdot 3^{2k-1} \\
&= 2(7m) + 7 \cdot 3^{2k-1} && \text{[by assumption]} \\
&= 7(2m + 3^{2k-1}) \quad \text{which is a multiple of 7.}
\end{aligned}
$$

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$2^{n+1} + 3^{2n-1}$ is a multiple of 7 for all $n \in \mathbb{Z}^+, n \geq 1$.

Q.E.D.

(d) Solve the equation $\log_x(2x-3)+\log_x(4)=2$. [5]

$\log_x(2x-3)+\log_x(4)=2$

$\log_x 4(2x-3)=2$

$\log_x(8x-12)=2$

Converting to exponential form gives:

$x^2=8x-12$

$x^2-8x+12=0$

$(x-2)(x-6)=0$

Either $x-2=0$ or $x-6=0$

$x=2$ $x=6$

Checking solutions: $x=2$ and $x=6$ are valid.

$\therefore x=2$ and $x=6$

Total: 20 marks

June 2018 – Question 2 – Solution

(a) A line, L_1, has equation $\boldsymbol{r} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and

a line, L_2, has equation $\boldsymbol{r} = \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ where α and β are scalar parameters.

Given that L_1 and L_2 meet at a point A, determine

(i) the coordinates of A [3]

We are given that L_1 and L_2 meet at a point A.

So, we have,

$$\begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} + \begin{pmatrix} \alpha \\ 2\alpha \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + \begin{pmatrix} \beta \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1+\alpha \\ 2+2\alpha \\ -4 \end{pmatrix} = \begin{pmatrix} \beta \\ 1 \\ -4 \end{pmatrix}$$

Equating corresponding entries gives:

$1 + \alpha = \beta$ $\rightarrow$ Equation 1

$2 + 2\alpha = 1$ $\rightarrow$ Equation 2

Consider Equation 1:

$2\alpha = 1 - 2$

$2\alpha = -1$

$\alpha = -\frac{1}{2}$

Substituting $\alpha = -\frac{1}{2}$ into Equation 1 gives:

$$1 + \left(-\frac{1}{2}\right) = \beta$$

$$\frac{1}{2} = \beta$$

Now, substituting $\beta = \frac{1}{2}$ into the equation for L_2 gives:

$$\boldsymbol{r} = \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} \\ 1 \\ -4 \end{pmatrix}$$

$\therefore$ The coordinates of A are $\left(\frac{1}{2}, 1, -4\right)$.

(ii) the angle between line L_1 and L_2, correct to 3 significant figures. [3]

Let $\boldsymbol{a} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

$$\boldsymbol{a} \cdot \boldsymbol{b} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= (1)(1) + (2)(0) + (0)(0)$$

$$= 1 + 0 + 0$$

$$= 1$$

$|\boldsymbol{a}| = \sqrt{(1)^2 + (2)^2 + (0)^2}$

$= \sqrt{1 + 4 + 0}$

$= \sqrt{5}$ units

$|\boldsymbol{b}| = \sqrt{(1)^2 + (0)^2 + (0)^2}$

$= \sqrt{1 + 0 + 0}$

$= \sqrt{1}$

$= 1$ unit

Now,

$\cos\theta = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$

$\cos\theta = \frac{(1)}{(\sqrt{5})(1)}$

$\cos\theta = \frac{1}{\sqrt{5}}$

$\theta = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$\theta = 63.4°$ (to 3 significant figures)

$\therefore$ The angle between line L_1 and L_2 is 63.4°.

(iii) Show that the vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is perpendicular to both L_1 and L_2. [3]

For perpendicular vectors, the dot product is zero.

Let $\boldsymbol{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

So, we have,

$$\boldsymbol{n} \cdot \boldsymbol{a} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= (0)(1) + (0)(2) + (1)(0)$$

$$= 0 + 0 + 0$$

$$= 0$$

And,

$$\boldsymbol{n} \cdot \boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= (0)(1) + (0)(0) + (1)(0)$$

$$= 0 + 0 + 0$$

$$= 0$$

Since both dot products are zero, then the vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is perpendicular to both L_1 and L_2.

Q.E.D.

(iv) Hence, determine the vector equation of the plane through the point $\begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$, which is perpendicular to the vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. [3]

Let $\boldsymbol{a} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The vector equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = (1)(0) + (2)(0) + (-4)(1)$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 + 0 - 4$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -4$$

$\therefore$ The vector equation of the plane is: $\boldsymbol{r} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -4$

(b) Solve the equation $\sin x + \cos 2x = 1$, where $0 \leq x \leq 2\pi$. [6]

$$\sin x + \cos 2x = 1$$

$$\sin x + 1 - 2\sin^2 x = 1$$

$$\sin x - 2\sin^2 x = 0$$

$$\sin x\,(1 - 2\sin x) = 0$$

Either $\sin x = 0$ or $1 - 2\sin x = 0$

$$-2\sin x = -1$$

$$\sin x = \frac{-1}{-2}$$

$$\sin x = \frac{1}{2}$$

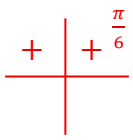
Consider,

$\sin x = 0$

$x = 0, \pi, 2\pi$

Consider,

$\sin x = \frac{1}{2}$



$x = \frac{\pi}{6}, \pi - \frac{\pi}{6}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

$\therefore x = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, 2\pi$ for $0 \leq x \leq 2\pi$.

(c) Determine the Cartesian equation of the curve with parametric equations

$$x = 1 + t^2\,,\ y = 1 + t$$ [2]

$x = 1 + t^2$ → Equation 1

$y = 1 + t$ → Equation 2

Rearranging Equation 2 gives:

$y - 1 = t$ → Equation 3

Substituting Equation 3 into Equation 1 gives:

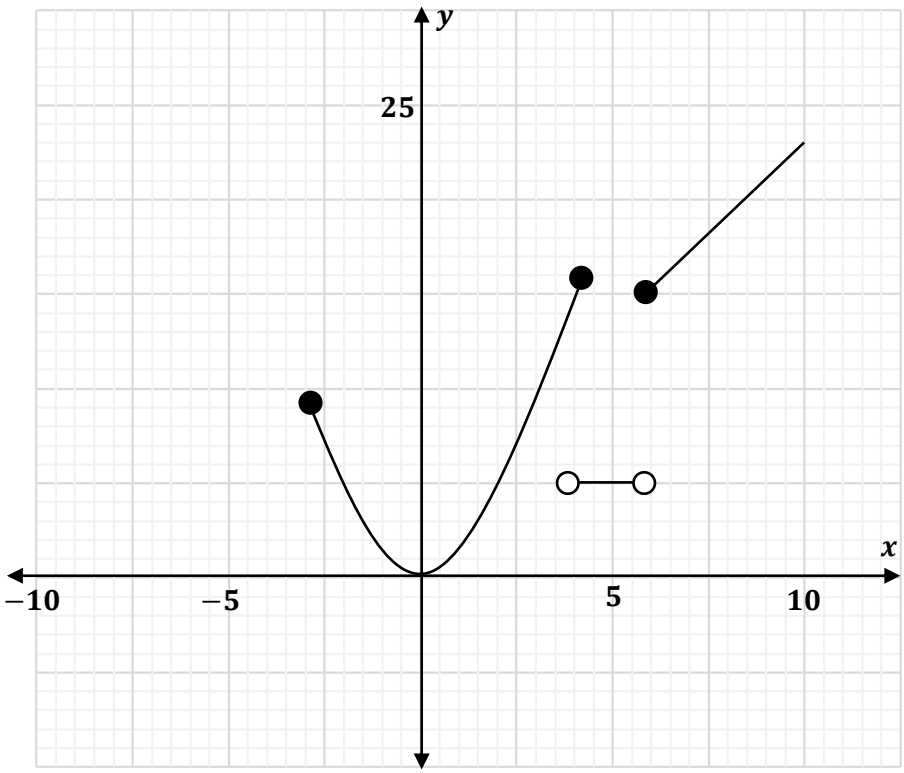
$x = 1 + (y - 1)^2$

∴ The Cartesian equation of the curve is: $x = 1 + (y - 1)^2$

Total: 20 marks

June 2018 – Question 3 – Solution

(a) The diagram below shows the graph of $y = f(x)$.



(i) Use the graph to find $f(6)$. [1]

From graph, $f(6) = 15$.

(ii) Determine whether the function is continuous at $x = 6$.

Justify your response. [2]

$\lim_{x \to 6^-} f(x) = 5$ and $\lim_{x \to 6^+} f(x) = 15$

Since $\lim_{x \to 6^-} f(x) \neq \lim_{x \to 6^+} f(x)$, then $\lim_{x \to 6} f(x)$ does not exist.

Since $\lim_{x \to 6} f(x)$ does not exist, then the function is not continuous at $x = 6$.

(b) Using **first principles**, differentiate $f(x) = \sin 2x$. [6]

$f(x) = \sin 2x$

Using first principles,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin[2(x+h)] - \sin 2x}{h}$$

$$= \lim_{h \to 0} \frac{\sin(2x+2h) - \sin 2x}{h}$$

$$= \lim_{h \to 0} \frac{2\cos\left(\frac{2x+2h+2x}{2}\right)\sin\left(\frac{2x+2h-2x}{2}\right)}{h}$$

$$= \lim_{h \to 0} \frac{2\cos(2x+h)\sin(h)}{h}$$

$$= 2\lim_{h \to 0} \cos(2x+h) \times \lim_{h \to 0} \frac{\sin h}{h}$$

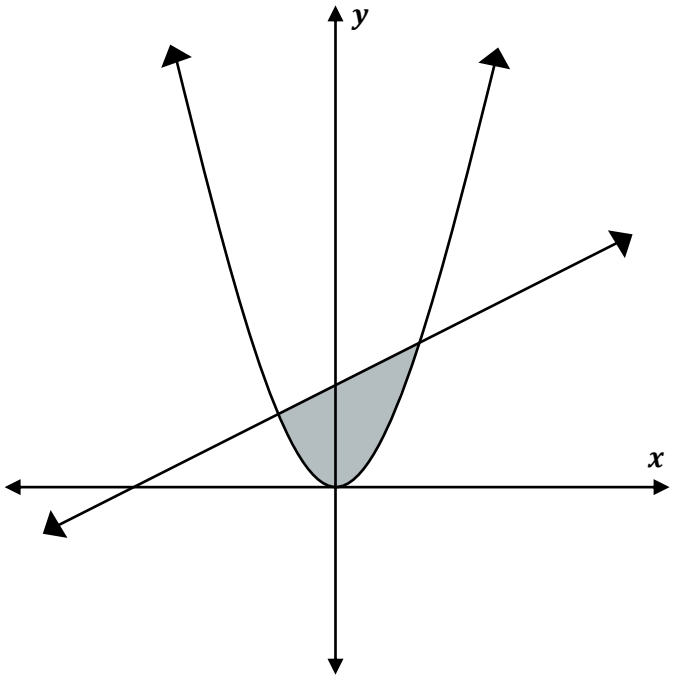
$$= 2\cos(2x) \times 1$$

$$= 2\cos 2x$$

$\therefore f'(x) = 2\cos 2x$

(c) The diagram below (not drawn to scale) shows the curve $y = x^2$ and the line

$y = x + 6$.



(i) Determine the points of intersection of the curve $y = x^2$ and the line

$y = x + 6$. [7]

$y = x^2$ → Equation 1

$y = x + 6$ → Equation 2

Equating both equations gives:

$$x^2 = x + 6$$

$$x^2 - x - 6 = 0$$

$$(x + 2)(x - 3) = 0$$

Either $x + 2 = 0$ or $x - 3 = 0$

$x = -2$ $x = 3$

When $x = -2$,

$y = (-2)^2$

$= 4$

When $x = 3$,

$y = (3)^2$

$= 9$

$\therefore$ The points of intersection are $(-2, 4)$ and $(3, 9)$.

(ii) Determine the area of the region bounded between the curve $y = x^2$ and the line $y = x + 6$. [4]

The line is $y = x + 6$.

Area under line $= \int_a^b y\,dx$

$= \int_{-2}^{3} (x + 6)\,dx$

$= \left[\frac{x^2}{2} + 6x\right]_{-2}^{3}$

$= \left[\frac{(3)^2}{2} + 6(3)\right] - \left[\frac{(-2)^2}{2} + 6(-2)\right]$

$= \left(\frac{9}{2} + 18\right) - \left(\frac{4}{2} - 12\right)$

$= \frac{45}{2} - (-10)$

$= \frac{45}{2} + 10$

$= \frac{65}{2}$ units^2

The curve is $y = x^2$.

$$\text{Area under curve} = \int_a^b y\,dx$$

$$= \int_{-2}^{3} x^2\,dx$$

$$= \left[\frac{x^3}{3}\right]_{-2}^{3}$$

$$= \left[\frac{(3)^3}{3}\right] - \left[\frac{(-2)^3}{3}\right]$$

$$= \frac{27}{3} - \left(-\frac{8}{3}\right)$$

$$= \frac{27}{3} + \frac{8}{3}$$

$$= \frac{35}{3}\ \text{units}^2$$

Now,

Area of shaded region = Area under line − Area under curve

$$= \frac{65}{2} - \frac{35}{3}$$

$$= \frac{125}{6}\ \text{units}^2$$

∴ The area of the region bounded between the curve and the line is $\frac{125}{6}$ units^2.

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.